



LECTURE: 26



ESD for Food Security

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Poor infrastructure

Poor infrastructure causes hunger by making it difficult sometimes impossible to transport food to areas of a country where there are shortages.

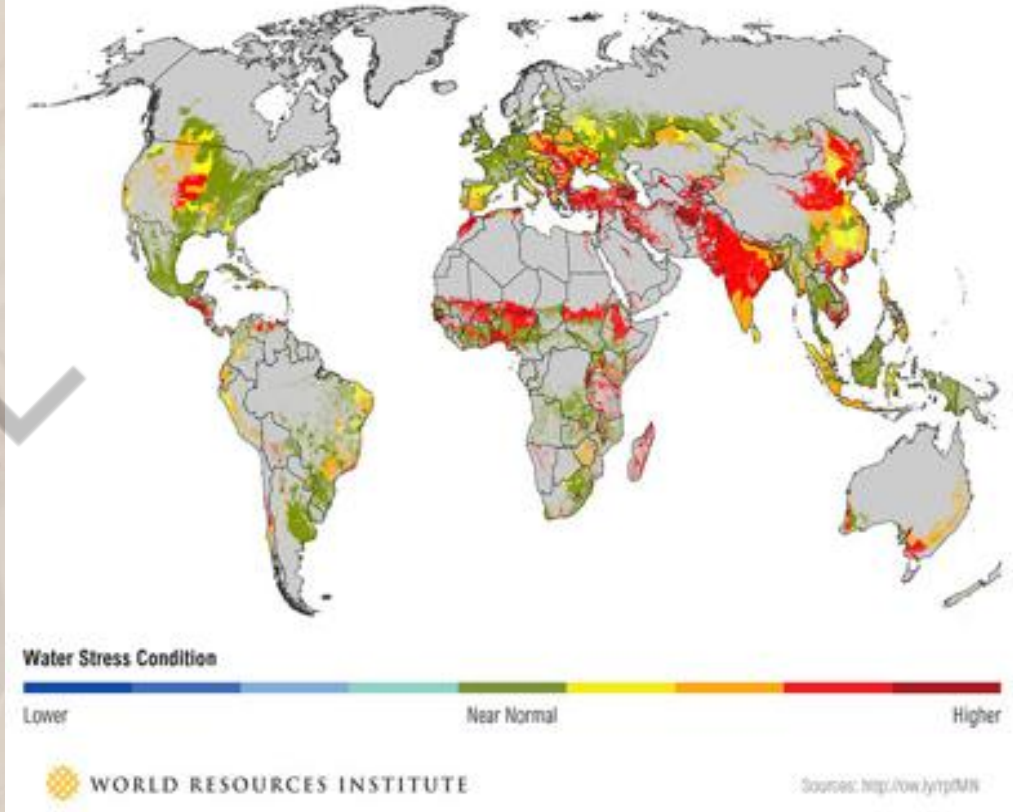
People have died of hunger in one region of a country while there was plenty of food in another region. The roads were so poor it was not possible to reach all who needed the food to survive.

Irrigation systems provide water for countries with unreliable or low rainfall. Irrigation can double crop yields, but it is expensive to put these systems in place. Water can be taken either from underground aquifers or directly from rivers. Both have environmental consequences.

Poor infrastructure contd...

- Crops need water to grow. **Irrigation infrastructure is unaffordable to most farmers** in developing countries. A **lack of water and sanitation infrastructure** are leading causes of hunger and malnutrition.
- Women and girls in developing countries spend hours each day fetching water because of a **lack of infrastructure**, pulling women away from other productive activities and girls out of school.

Water stress will increase in many agricultural areas by 2025 due to growing water use and higher temperatures (based on IPCC scenario A1B)

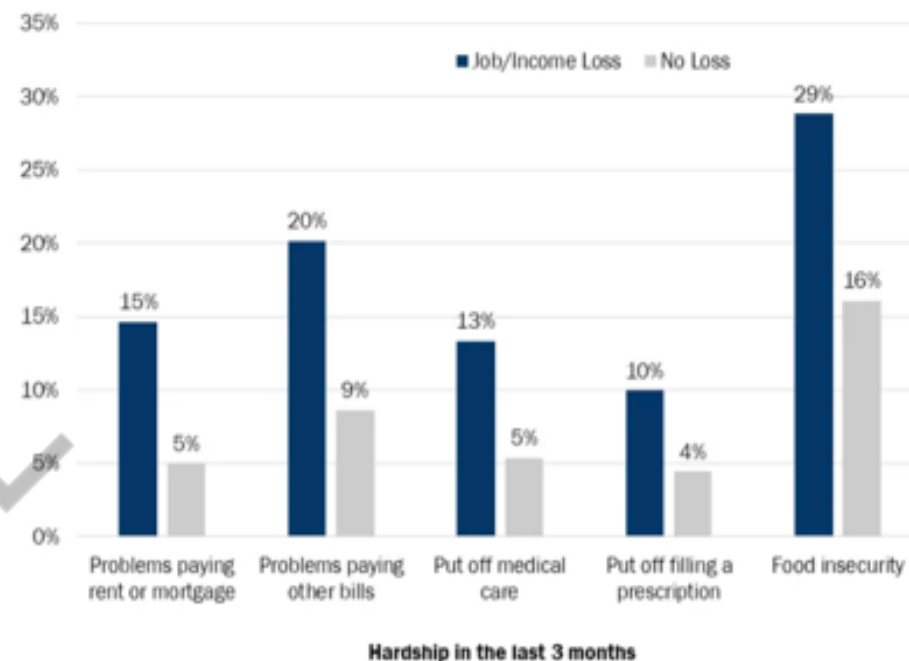




Job Instability

- ✓Hunger rates rise when the **national or local economy is in a slump**. People **lose jobs** and **cannot find work**. Once the economy improves some people continue to struggle to find work.
- ✓In **single-parent families**, the parent may **not** be **able to** take a **job or work** enough hours because of no childcare options.
- ✓In the United States and other high-income countries, hunger is mainly caused by poverty that results from a **lack of jobs** or because jobs pay too little.

Figure 4. Hardships and COVID-19-related job or income loss



Source: COVID-19 Survey, Wave 1 (April 22–May 12, 2020), Social Policy Institute.

BROOKINGS



War and Conflict

Hunger is both a cause and effect of **war and conflict**. Wide-scale poverty and hunger lead to frustration and resentment with governments that appear to ignore hungry people's plight.

The poorest members of society suffer the worst during war and conflict. **Homes are destroyed** and communities of **people are displaced**. Peace when it comes is often tenuous. The physical **infrastructure** needed for reconstruction is **damaged** and might even be **destroyed**.

Unstable Market: Any fluctuation that pushes food prices up in the market creates additional hardship.



- Farmers in Developed Countries can afford **pesticides**, whereas most farmers in low income developing countries (LIDCs) cannot afford them.
- **Disease** closely **related to issues of poverty and food insecurity** are problems of disease due to malnutrition, scarcity of water for drinking, poor sanitation, pollution, and inadequate shelter; those are often compounded by the spread of infectious diseases such as malaria, cholera, tuberculosis and HIV/AIDS.
- Large differences occur in the responses of human societies to diseases, reflecting **vast inequalities in health care** spending and in funding for pharmaceutical and medical research



- Technology can overcome temperature, water and nutrient deficiencies in the form of greenhouses, irrigation and fertilizers. This can incur an **economic or environmental cost**. Developed Countries import food from across the globe, all year round.
- **Discrimination**: Progress against hunger and poverty seldom happens without economic growth in countries, but economic growth alone does not ensure that prosperity is broadly shared. Every country, regardless of its wealth, has discrimination woven into its social fabric. Disadvantaged groups tend to be left the furthest behind.
- **Discrimination is why women farmers in developing countries labor with fewer productive resources** than their male counterparts, why women in all sectors of the economy earn less than men, and why girls are pulled out of school to work or to marry.



- **Nutritional Quality:** All people who are hungry are malnourished. They are not getting enough protein, so they lose weight and in severe cases their bodies begin wasting.
- Another form of **malnutrition** is known as "**hidden hunger**," and it has more to do with the quality of food than the quantity
- **Infants and young children are most vulnerable** to the harmful effects of hidden hunger.
- **Loss of farmland:** The growth of the biofuel market is taking up valuable farmland which is then not used for food. Intensive farming often leads to **a vicious cycle of exhaustion of soil fertility and decline of agricultural yields.**
- **Desertification** is the process by which fertile land becomes desert, typically as a **result of drought, deforestation, or inappropriate agriculture.**



Suggested Solutions to Cut Poverty

➤ Given the rapidly growing population, resource constraints and climate change concerns, accompanied by the impacts of the COVID-19 pandemic; it is imperative for **India to make food security a core policy priority. A few policy changes are essential to improve India's hunger situation:**

1. Create jobs
2. Raise the minimum wage
3. Increase the Earned Income Tax Credit for childless workers
4. Support pay equity
5. Provide paid leave and paid sick days
6. Establish work schedules that work
7. Invest in affordable, high-quality child care and early education
8. Expand Medical aid
9. Reform the criminal justice system and enact policies that support successful re-entry
10. Do no harm



11.Social protection in the form of cash transfers will eliminate hunger immediately, it noted, adding that public investments in rural areas could be through small-scale irrigation projects, creating infrastructure such as roads, food processing and stronger institutional support like access to credit and markets.

12.Activities such as upgradation of rural infrastructure, training of farmers in post-harvest practices to minimize losses, integrating small scale enterprises into value chains, organizing smallholder farmers into farmer producer organizations, customized financial services, investment in agricultural research, and last-mile marketing channels should also come into action.

13.The 'One Nation, One Ration Card' across the country is a great move in this direction and no hurdle should allow it to stop its universalization soon so that the PDS can be accessed at any geographical location in the country.



- Home delivery of meals and cooking material to the beneficiaries of the supplementary nutrition programme under the **ICDS and MDM** for as long as Anganwadis (childcare centres) and schools remain closed.
- Nutritious food should be provided to the vulnerable section of the population, especially during the pandemic and its aftermath, as this can help avoid heightening food insecurity.
- **Sustainable agriculture practices** — such as crop rotation, mixed cropping with pulses, using **biofertilizers**, limiting the use of pesticides or fertilizers, and integrated pest management — must be encouraged and promoted.



LECTURE: 27

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- **Drip irrigation** and solar panels for electricity generation will lead to the **conservation of natural resources**.
- **Investment is needed** for the development and distribution of **climate-resilient crops** that can handle temperature variation and precipitation fluctuations.
- It is crucial to reconsider **providing subsidies for certain crops** as they are having an adverse impact on the natural resources.
- Given the risks associated with farm incomes, the **government must continue rendering support in the form of agricultural credit, subsidies on inputs and disaster relief along with micro-insurance** for crop, weather and livestock.



- To counter the economic impacts of the pandemic, including job losses and reverse migration, the **MGNREGA programme** must be scaled up. In the past, MGNREGA has increased rural wages, reduced gender wage gaps, enabled better access to food and reduced distress migration from rural areas.
- Post the pandemic, the programme has the potential to cater to the health, nutrition and livelihood needs of the migrants who have returned home. MGNREGA has also made an important contribution to child wellbeing, through the reduction of hunger and improvement of health and education, and it **must continue to be promoted**.

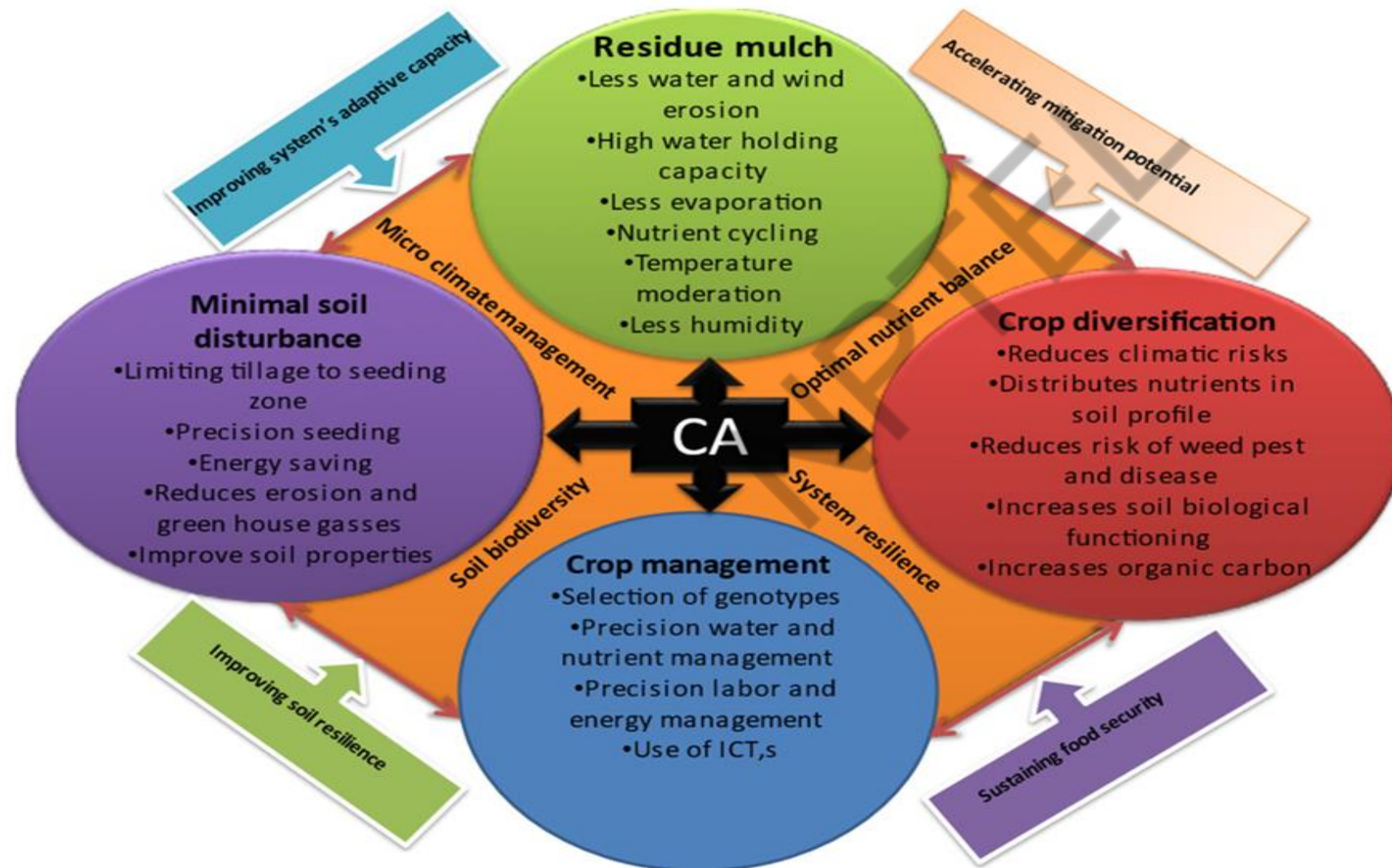


➤ Increased budget, technology, research& development

- In the 2021-22 Union Budget, funding allocations for the health sector have been nearly doubled due to the pandemic. The Budget also allocated more funds towards agricultural credit, and for the promotion of animal husbandry, dairy and fisheries sector.
- But merely allocating funds will not be enough for a nation that is home to about a third of the world's malnourished children.
- Farmers' access to the latest research can help in improving seed quality, decreasing pest and disease problems, increasing crop sustainability, reducing irrigation problems, lessening incidents of soil erosion, and increasing productivity to feed a burgeoning population.



Source: MDPI





Climate Change and Agriculture

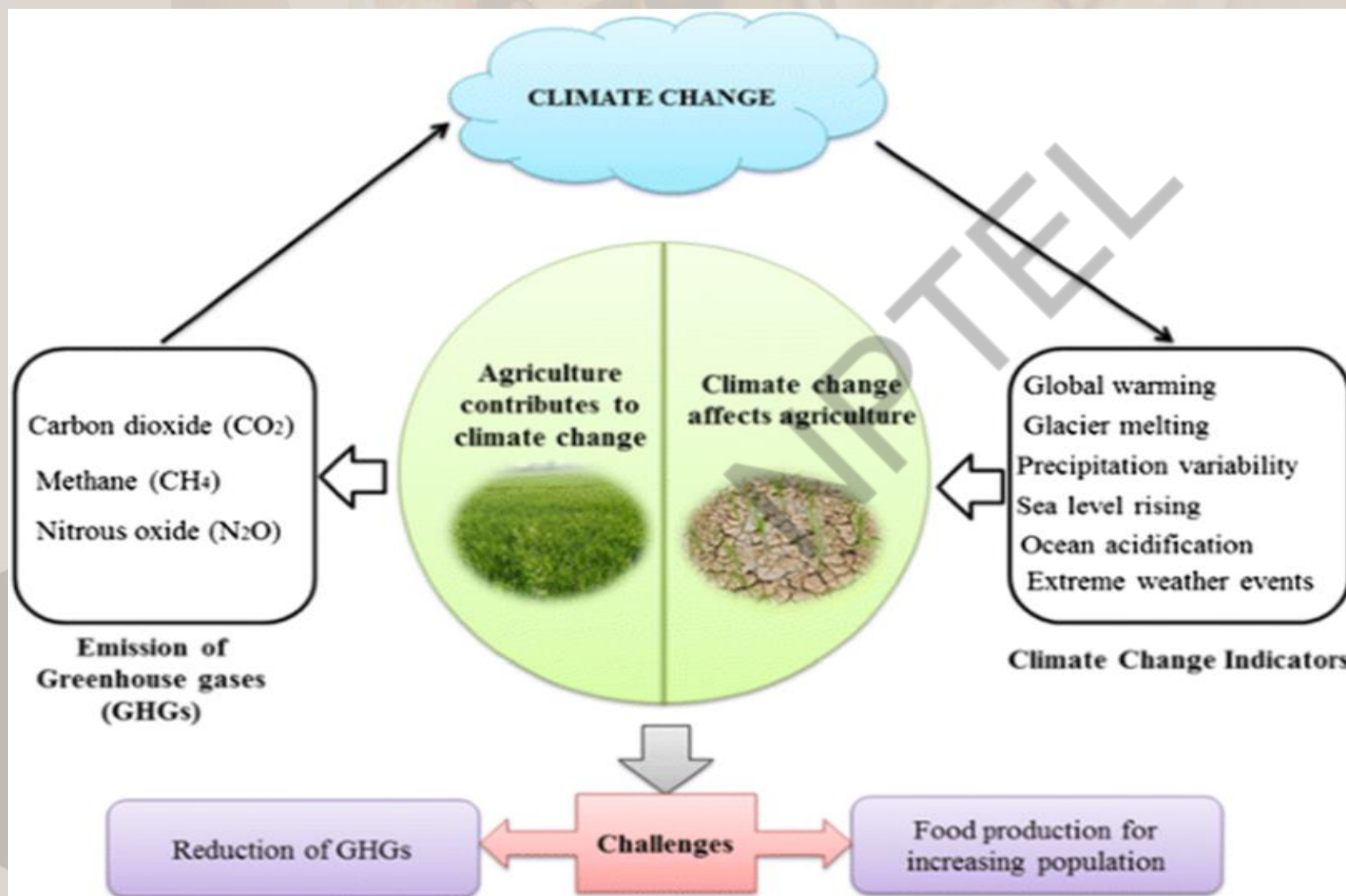
- ✓The **most serious long-term challenge** facing Indian agriculture is **global warming and Climate Change**.
- ✓It is clear that India will suffer a major setback in wheat production as global warming increases. Wheat productivity, which is critically dependent on cold night temperatures, will suffer as temperatures rise.

What can be done?

- ✓**Wheat displacement** – with millets, tubers like potato, yams and cassava. Start multiplying seeds and other planting material right away so that there is enough planting material for the new crops when the season begins.
- ✓**Diversify food basket** – bring forgotten and neglected foods like millets, yams, leafy green vegetables and other underutilized crops into cultivation and onto the market.

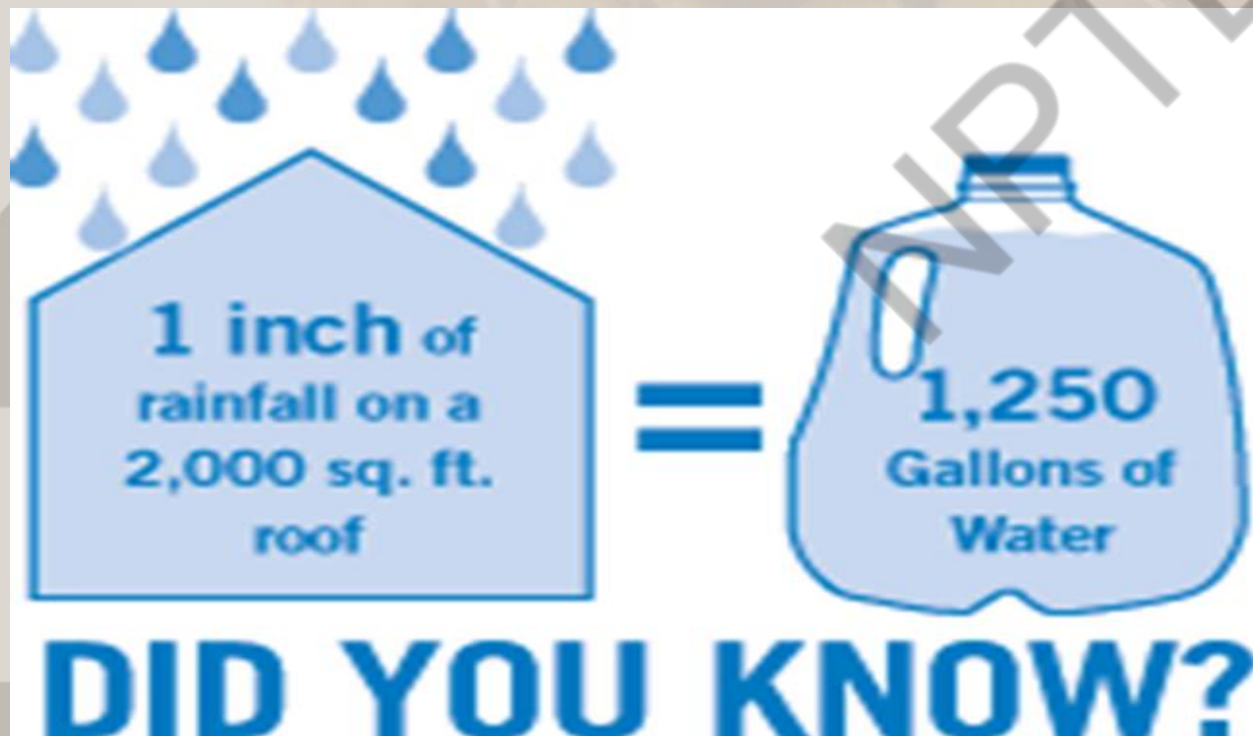


Source: Traditional agriculture: a climate-smart approach for sustainable food production, Rinku Singh & G. S. Singh





- **Watersheds are important to recharge groundwater** but to increase agriculture production in the shorter term, the farmers need to access the water for the winter crop.





- **Role of the private sector**

- The **private sector can play an important role** in complementing the achievement of a higher level of food security in a country, at least during peacetime.
- to establish **essential regulatory frameworks** under which the private sector can work in harmony with public efforts in improving the country's food security and under which unwanted activities are forbidden or penalized.
- This requires the **presence of strong institutions** in the country, especially **strong government and economic institutions.**



Walmart, hunger relief organizations and food banks. Photo by: Walmart / CC BY



- **Supporting harmonious global trading institutions**
- Most countries in Asia, especially in East Asia and Southeast Asia, will increasingly rely on imports to maintain their domestic food supplies. As such, **stable food imports** will become more important. Such import stability can be best protected by world peace; thus making contributions to **long-term world peace** would be most rewarding.
- Even in a peaceful world, **trade barriers exist**, and trade disruptions occur; it would be most beneficial for India to contribute to the **establishment and maintenance of harmonious global trading institutions**, through which trade disruptions can be minimized.
- **Empowering women TO PROMOTE FOOD SECURITY**



✓ **Educating women** about how to **empower** her household. It is imperative because women suffer from malnutrition, and women are closer to children who also suffer negatively from food insecurity.

✓ So, to teach women and give them an **education about food and nutrition** would be suitable for food household security.

✓ "**Mother's education** is positively associated with better children education, health, and nutrition outcomes" (Sidh, 2011).





- **Home gardening** has been proven as a reliable source to supplement food and nutritional security at the household level
- This program, especially in India, could be helpful if combined with a **government program that giving subsidy** to its people but still in terms of staple food not cover other Nutritional food yet.
- So, when the government can focus firstly on giving aid to its people to obtain food in terms of access, the **community can help to empower themselves to provide nutritious food through home gardening.** Combining these programs, hopefully, can solve food insecurity.





• REDUCING FOOD WASTE

- The government, on its part should make laws that penalize companies for wasting food within their supply chain and encourage repurposing and recycling of food items.
- For example, since 2016, France has been fining grocery stores for throwing away edible food.
- 'Disposing is cheaper than using or reusing' attitude should be changed gradually.
- Private companies should invest to bring out potential solutions to minimize food wastage.
- Companies can influence consumer behavior by advocating- 'Buy one get one free—later' programme wherein customers can buy certain food items and take the free product later when they actually need it.





- **AI to avoid food wastage**
- **US-based Winnow uses computer vision to help commercial kitchens** around the world understand how food is being wasted, measuring the amounts discarded using smart sensors attached to waste bins. The app also proposes strategies to prevent waste while saving money.
- The Berlin-based start-up solution **Plantix** aims to stop good crops from being wasted when a farmer presumes them to be damaged. The so-called "mobile crop doctor" uses image recognition to help farmers diagnose any issues with their crops, thereby preventing edible food from being lost.
- **Wasteless**, meanwhile, uses **artificial intelligence (AI)** to maintain dynamic pricing on items with upcoming expiration dates. The objective is to help supermarkets and online grocery stores in Europe and the US reduce food waste and extend the value of their perishable food items.



LECTURE: 28



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2

ZERO HUNGER



ZERO HUNGER

Challenges

We live in a world where
1 IN 9 PEOPLE IS UNDERNOURISHED



75% of crop diversity has been lost to **MONOCROP CULTURE** making crop more **VULNERABLE TO PEST OUTBREAKS AND DISEASES**



SMALLHOLDER FARMERS provide 80% of the food consumed in the developing world. Mostly rainfed their agriculture is particularly **VULNERABLE TO DROUGHTS AND FLOODS**



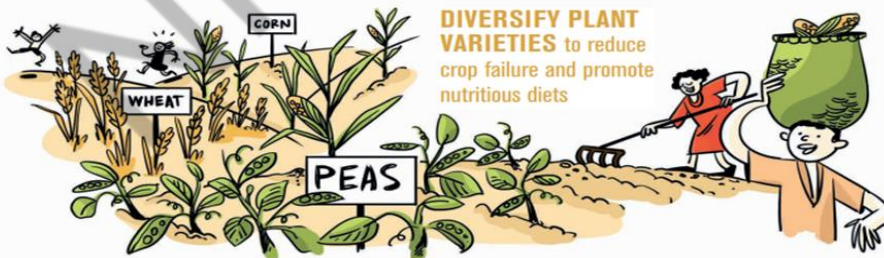
Solutions



INCREASE AGRICULTURAL PRODUCTIVITY and **ENSURE EQUITABLE ACCESS** to food for all



DIVERSIFY PLANT VARIETIES to reduce crop failure and promote nutritious diets

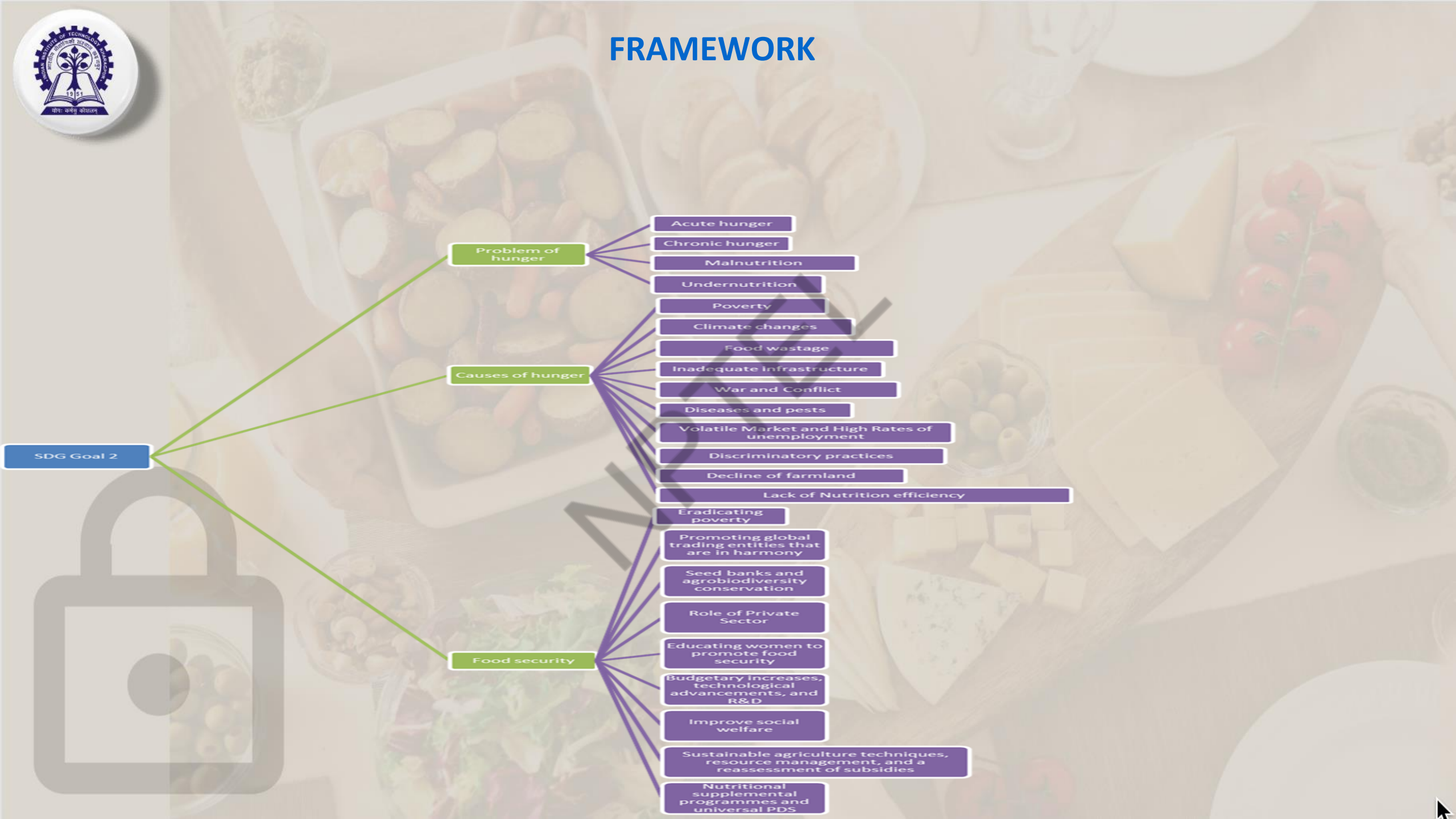


INVEST IN SMALLHOLDER FARMERS especially **WOMEN** to increase food security and improve nutrition for the poorest





FRAMEWORK





➤ Suggestions

Global experience shows that with the **right public policies focusing on agriculture, improved sanitation, and women's education**, one can have much better health and well-being for its citizens, especially children.

In China, it was agriculture and economic growth that significantly reduced the rates of stunting and wasting among the population and lifted millions of people out of hunger, poverty and malnutrition.

According to the Global Nutrition Report, 2016, at the present rates of decline, **India will achieve the current stunting rates of China by 2055. India can certainly do better, but only if it focuses on this issue.**



Research Findings

- 1. The world population continues to grow. Food production needs to rise by 50% by the year 2030 to meet the needs of population growth and changes in consumption patterns. The tendency is clear: the more countries have economic development, the more they consume animal protein. However, meat production is a major source of greenhouse gases and the typical land use attached to it is generally inefficient and prone to cause soil degradation. Resource use related to agriculture and food production is highly unsustainable. Agriculture is still the most common occupation globally. Thus attention is needed in linking education and food security.
- 2. Today's agricultural and trade practices are failing to feed the poor and have detrimental effects on the environment. Current cultivation of 'cash crops' such as cotton and coffee for export purposes contribute to the prevalence of food insecurity and create a strain on the ecosystem. These practices need to be re-considered with the holistic approach to education and development



- 3. In 2017, the Agriculture, Fisheries and Conservation Department of Hong Kong Government revealed that local production accounted for only 1.7 percent of fresh vegetables. **Such over-dependence on food import from foreign countries showcases the need to prevent further loss of the existing farmland** and, instead, expand urban agriculture in the city. In relation to that, Sustainable Development Goal (SDG) 2 highlights the significance of sustainable agriculture and SDG 4.7 insists that, **by 2030, all learners should acquire the knowledge and skills** they need to achieve sustainable development (UNESCO, 2016). According to the Food and Agriculture Organization of the United Nation (2018), farmer field school approach was applied in more than 100 countries to educate youth on agro-ecosystem agriculture techniques to support the work towards SDGs, and, in particular, SDGs 2 and 4.7.



- 4. The Sustainability Office at HKU has organized a range of **programs to promote sustainable development on campus, including 'Green Connections' and 'Edible Spaces'**. Apart from HKU, almost all universities in Hong Kong have a Sustainability Office or an organic garden where workshops or activities are organized. Chinese University of Hong Kong, for example, has had a rooftop farm since 2012 for students and staff from the School of Architecture to grow crops. In addition, an associate professor and experienced organic farmer Mr. Fung has a rooftop cultivation project and offers a weekly workshop on urban farming.





- 5. Urban farming education could be considered an effective and appropriate approach to ESD in Hong Kong, which is a very compact city. However, more efforts should be made to meet the standards of ESD 2, along with strengthening the ESD 1 approach. HKU has offered a well-organized urban farming program for its students regarding **up-to-date knowledge in organic rooftop farming and hands-on practices which helps to achieve SDG 4.7**. Nevertheless, the program should be improved to include the following aspects: observational skills, vocabulary enrichment, content on moral, social, and economic implications, tasting experience, and lifelong learning promotion (Francis et al., 2017; Polat, 2015).





- 6. The world population continues to grow. Food production needs to rise by 50% by the year 2030 to meet the needs of population growth and changes in consumption patterns. The tendency is clear: the more countries have economic development, the more they consume animal protein. However, meat production is a major source of greenhouse gases and the typical land use attached to it is generally inefficient and prone to cause soil degradation. Resource use related to agriculture and food production is highly unsustainable. Agriculture is still the most common occupation globally. Thus **attention is needed in linking education and food security.**
- 7. **Today's agricultural and trade practices are failing to feed the poor and have detrimental effects on the environment.** Current cultivation of 'cash crops' such as cotton and coffee for export purposes contribute to the prevalence of food insecurity and create a strain on the ecosystem.



- 8. Knowledge of the current state can lead to empowerment ; people need to take responsibility for their actions, which is one of the primary aims of ESD.
- The thrusts of the **ESD** are providing and improving **quality basic education, re-orienting existing education programs, building public understanding and awareness and providing practical training.**
 - **ESD brings sustainable issues to the forefront**, and uses that perspective to address educational change which enables everyone to adapt their behaviors and make informed decisions that can contribute to a sustainable future.
 - **ESD aims to provide all learners with support to develop skills to adapt to changing food security needs in their local context** - for example, to seek out innovative farming methods and change consumption patterns.



- 9. Current practices need to be re-considered with the holistic approach to education and development that ESD advocates. It seeks to enable individuals to view issues of food security as connected to their personal choices. For example, by incorporating ESD in vocational education programs and agricultural practices, farmers can be transformed into 'farmers of tomorrow' who are better able to safeguard our ecosystems and preserve them for future generations. The way to this is through exchanges in expertise between indigenous knowledge and future farmers. Community knowledge of local contexts must be integrated into sustainable approaches in order to address local needs and opportunities for farmers in maximizing agricultural production



- **10. Food production index** and by implication food security is a major concern of the government. **Government expenditure on agriculture** and exchange rate were stationary at level while food production index, **inflation rate and interest rate** were stationary at first difference. There was a single co-integrating equation showing that there was a long run **relationship among variables**.
- **In the short run, only government expenditure positively affected food production index significantly.** In the long run government expenditure on agriculture, inflation rate and interest rate negatively affected food production index and exchange rate positively affected food production index. Therefore,
 - **Government budget allocation to agriculture should meet up** with the Maputo agreement, to ensure the necessary increase of food production;
 - Interest rate should be regulated by the apex bank.



A framework for modelling the complexities of food and water security under globalization

The framework sets out a method **to capture regional and sectoral interdependencies and cross-scale feedbacks within the global food system that contribute to emergent water use patterns**. The framework integrates aspects of existing models and approaches in the fields of hydrology and integrated assessment modelling. The core of **the framework is a multi-agent network of city agents connected by infrastructural trade networks**. Agents receive socio-economic and environmental constraint information from integrated assessment models and hydrological models respectively and simulate complex, socio-environmental dynamics that operate within those constraints. The emergent changes in food and water resources are aggregated and fed back to the original models with minimal modification of the structure of those models. It is our conviction that **the framework presented can form the basis for a new wave of decision tools that capture complex socio-environmental change within our globalized world**. In doing so they will contribute to illuminating pathways towards a sustainable future for humans, ecosystems and the water they share.

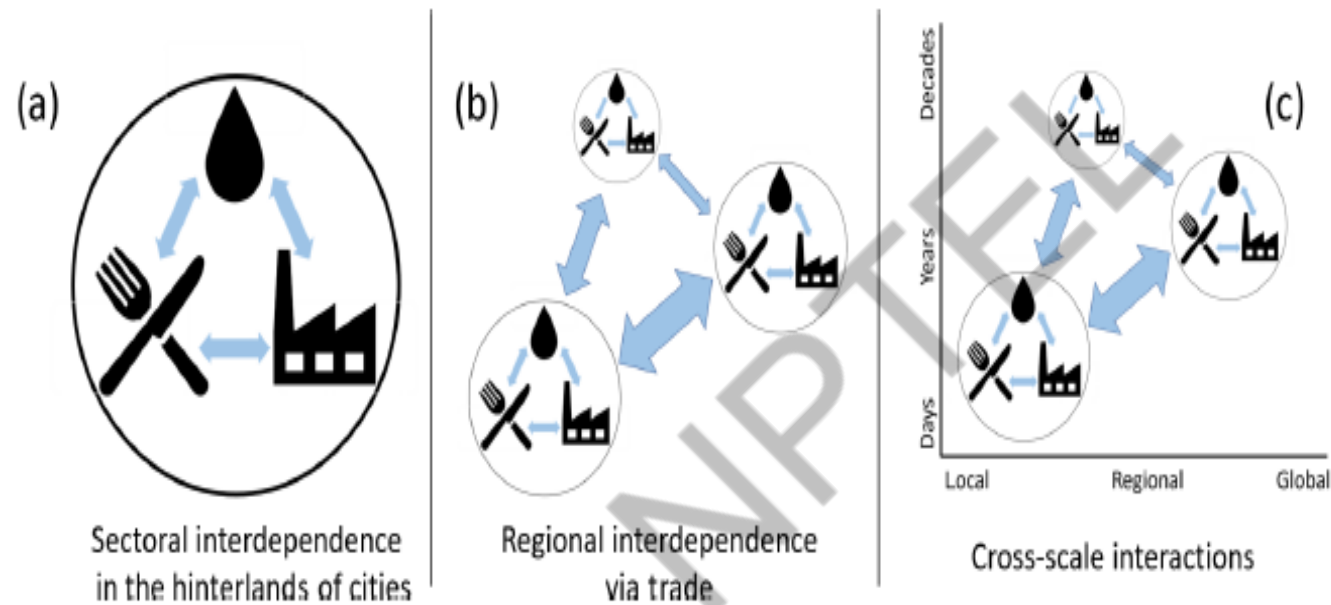


Figure 1. Complex dynamics within the global food system that lead to emergent water use patterns. (a) Sectoral interdependence in water resources within the food–water–energy nexus; (b) regional interdependence in water resources owing to the importance of trade for food security; (c) cross-scale feedbacks whereby fine-scale interactions bring about emergent higher-level change. Higher-level, emergent changes may feed back on the fine scale (Sivapalan and Blöschl, 2015).

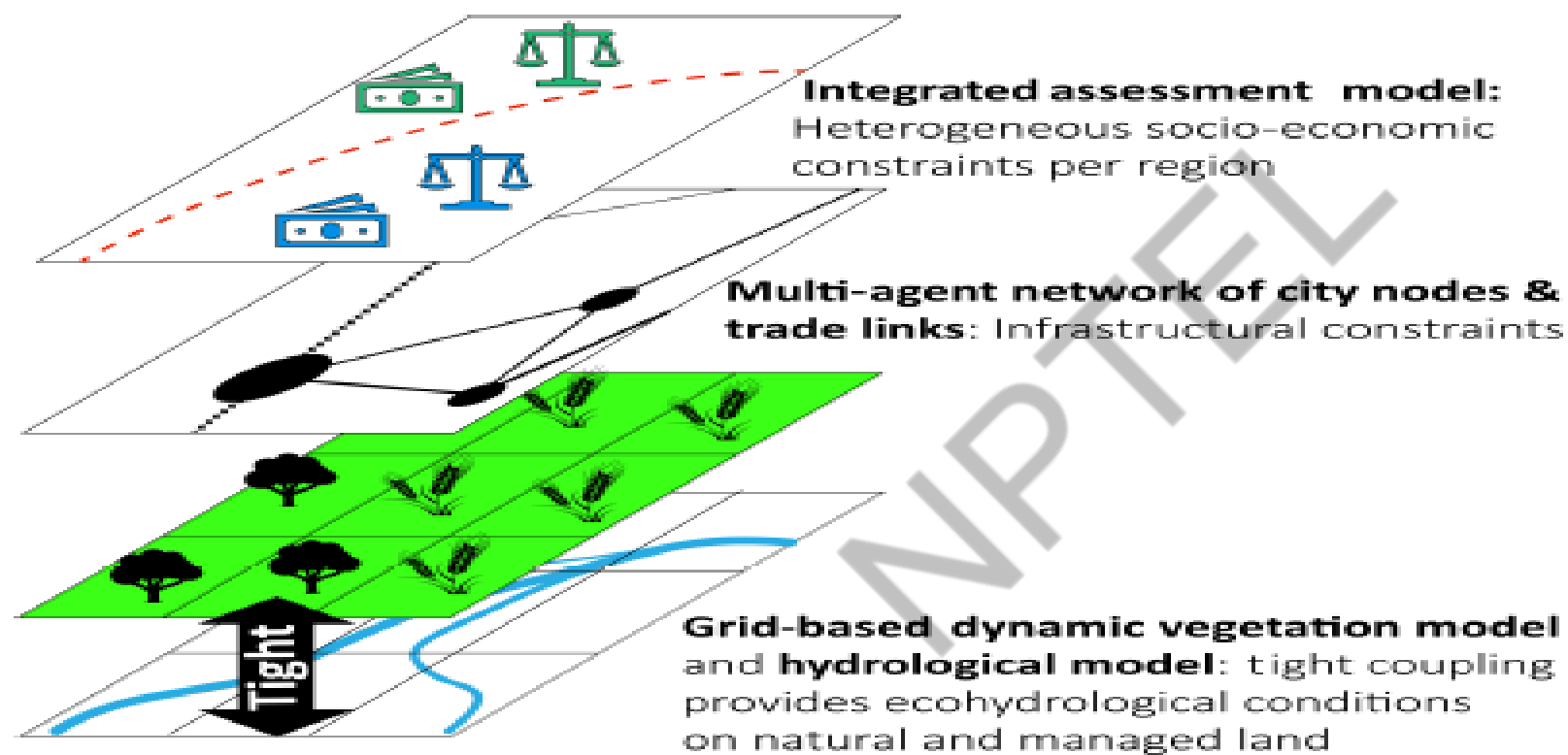
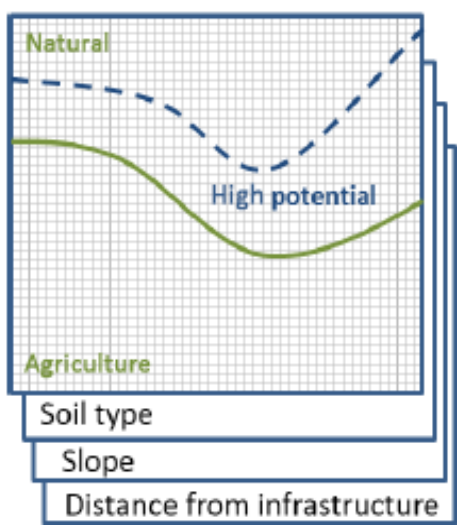


Figure 3. City agent attributes. In our framework, city agents receive ecohydrological conditions from a tight coupling between a dynamic vegetation model and hydrological model. They receive socio-economic constraint information from an integrated assessment model. Ecohydrological conditions on natural and managed land determine food production potential and associated water use.



(a) Expansion potential



(b) Intensification potential

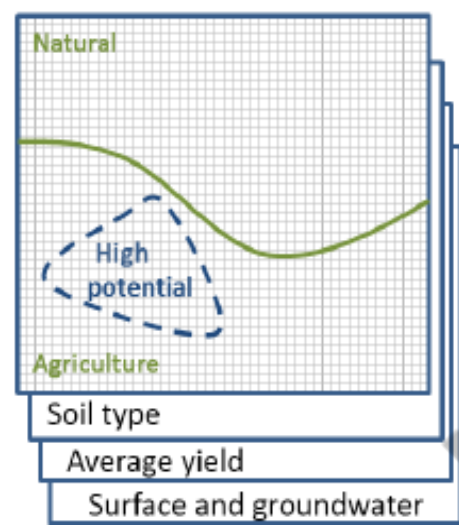


Figure 5. Food production decisions. City agents decide on agricultural expansion (a) or intensification (b) based on demand and spatially explicit crop production potential in their hinterlands.

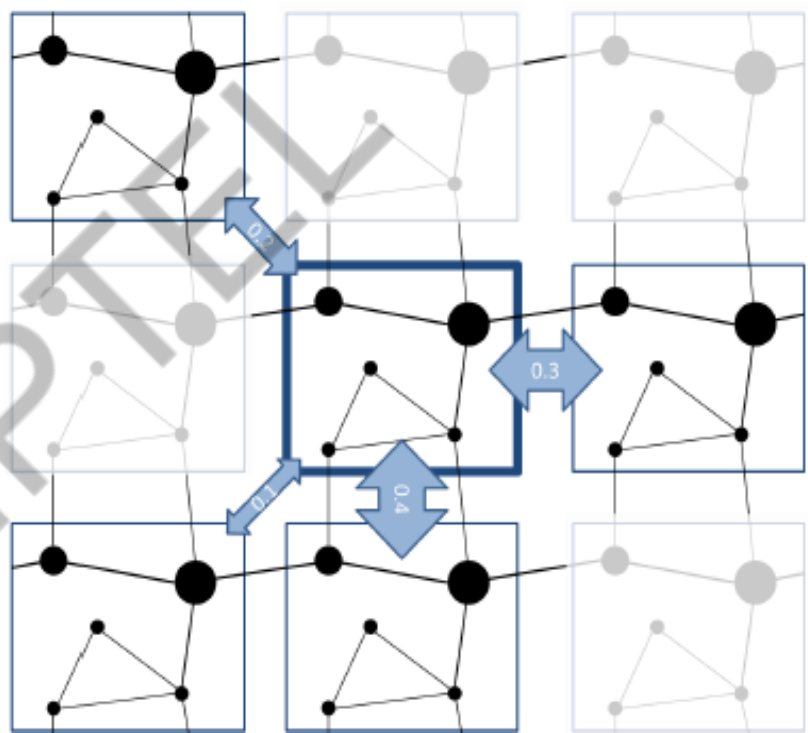


Figure 6. Virtual water trade network. The trade network is hierarchical, containing an upper-level network with edge weights cap-

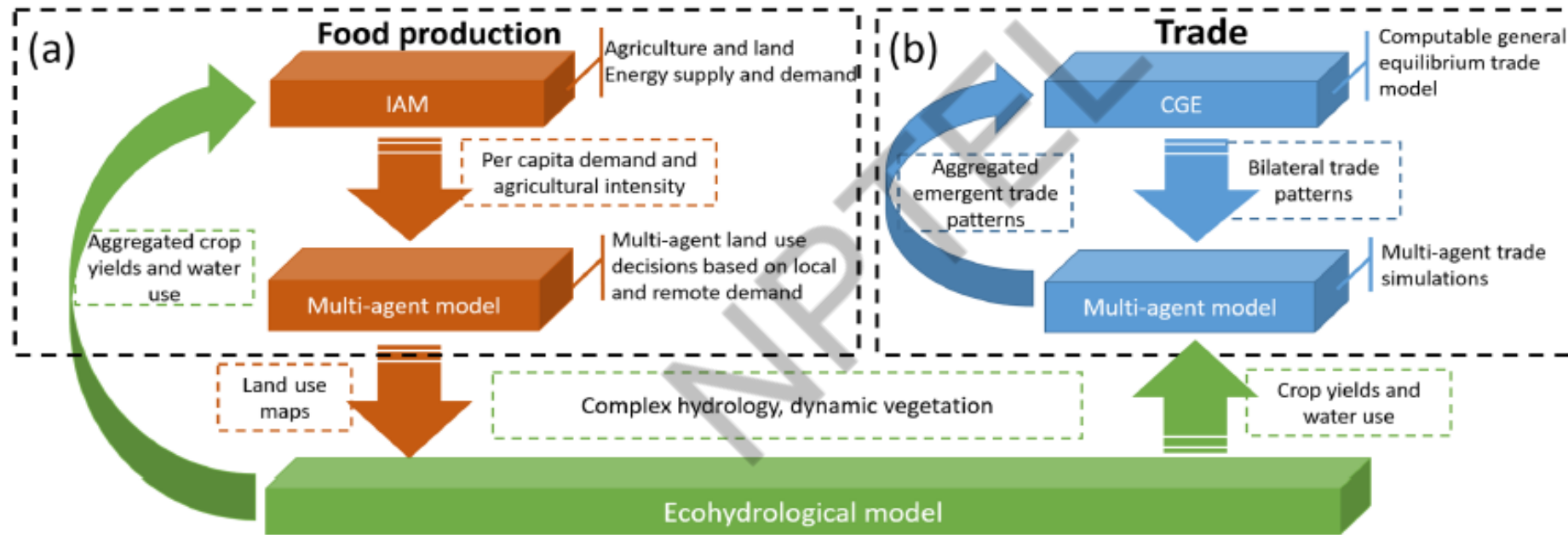


Figure 7. Modelling framework workflow. (a) The multi-agent network sits at the interface between an IAM and an ecohydrological model. The IAM calculates per capita food demand on a regional scale, which is converted into spatially explicit demand using population maps. The IAM also calculates agricultural intensity constraints. Based on these constraints and demand, agents make food production decisions that change ecohydrological conditions. The emergent crop production and water use patterns are aggregated and feed back to the IAM. (b) A CGE trade model provides edge weights for the upper-level trade network. Based on these bilateral trade constraints and demand, virtual water trade among city agents is simulated among agents via the infrastructural trade network. The emergent aggregated trade patterns feed back to the CGE as input for the next simulation year.

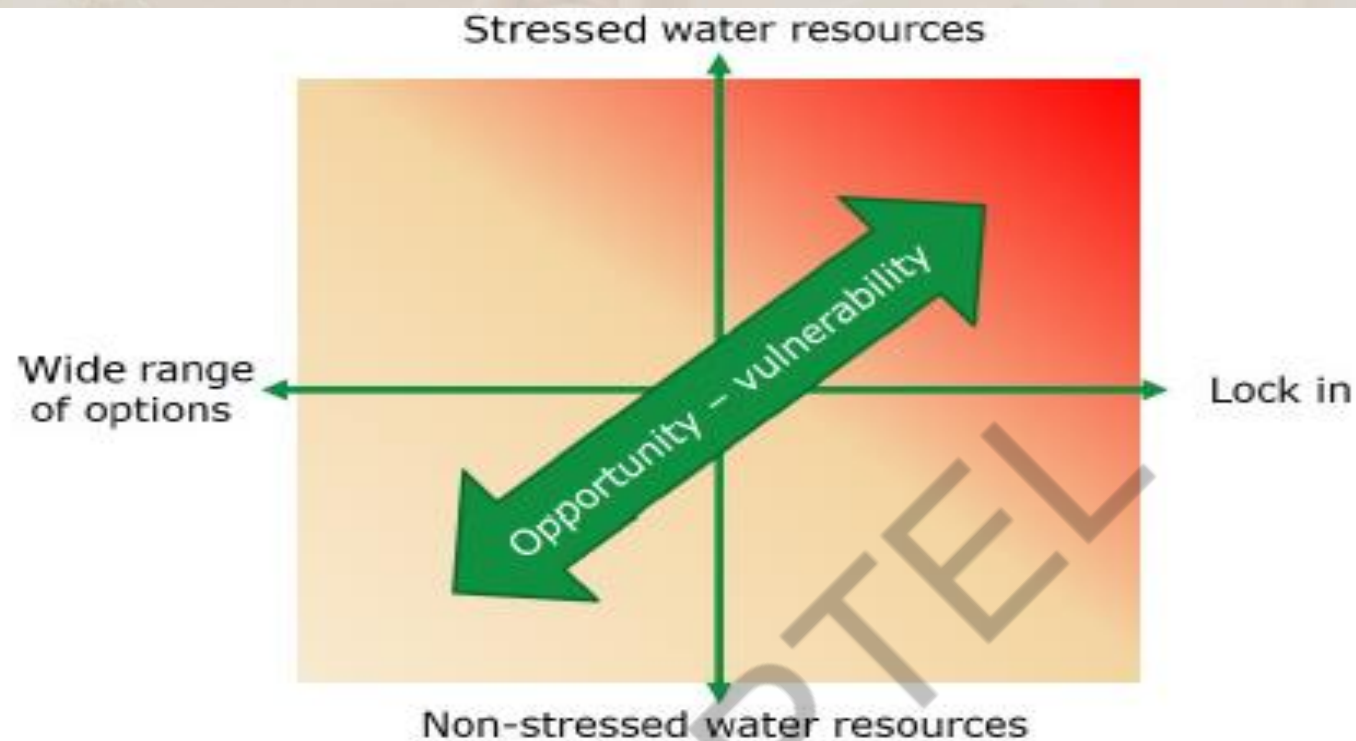
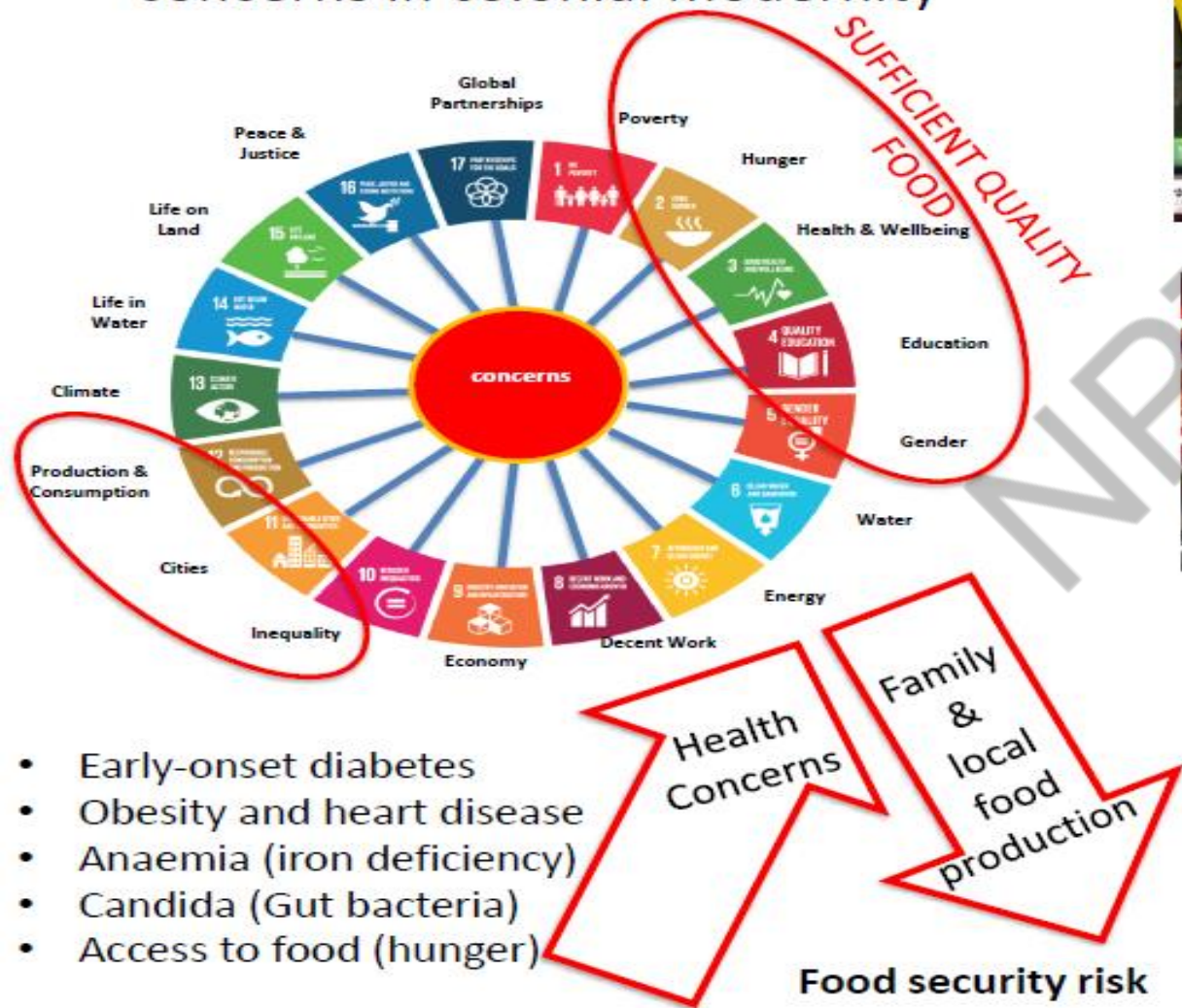


Figure 8. The solution space reveals how constrained regions or sectors are along their current water use pathway. The solution space emerges from the multi-agent simulations. The solution space may be explored to find pathways towards sustainable water futures that take account of regional and sectoral interdependencies and cross-scale feedbacks. Those regions or sectors that are locked into unsustainable or vulnerable water use pathways should be prioritised for intervention.



PRESENT: Sustainability and health concerns in colonial modernity



Cultured milk (curds)



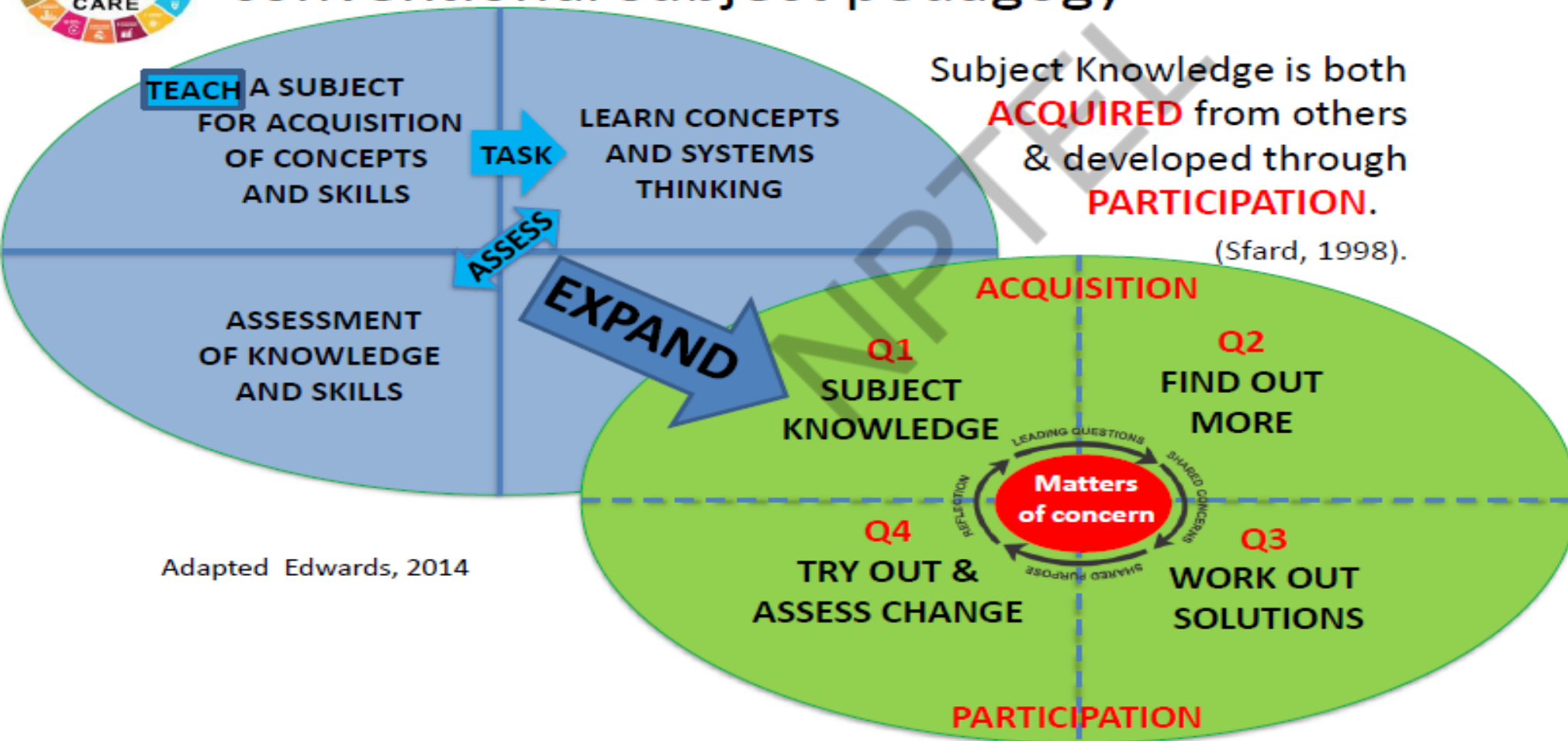
Leafy green vegetables



- Early-onset diabetes
- Obesity and heart disease
- Anaemia (iron deficiency)
- Candida (Gut bacteria)
- Access to food (hunger)

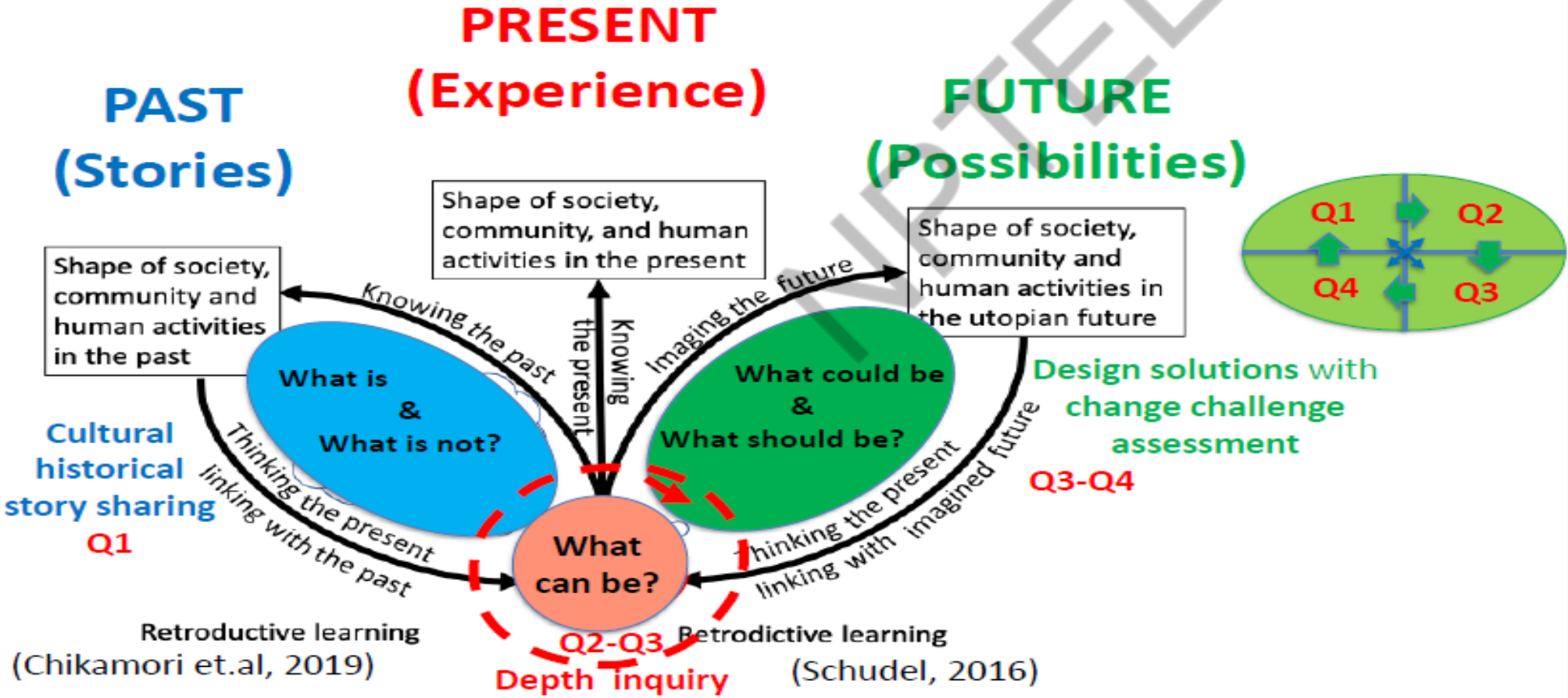


FUTURE: An **action learning expansion** of conventional subject pedagogy





Critical Realist Learning across **PAST** and **PRESENT** to re-imagine a sustainability **FUTURE**





CULTURED MILK (*Amasi*)



CULT

1 Tune-in
How were things in the past?



1 Tune-in
How were things in the past?



2 How are things today?



4 How can we make things more sustainable together?

4 How can we make things more sustainable together?



Shop-bought
Home-Made?
or
Both

Shop-bought



Home-Made?
or
Both



3 What does this mean for us today?



Touch the past with our memories, feel the future flying on the wings of imagination
(Mhlope, 2021)

Touch the past with our memories, feel the future flying on the wings of imagination
(Mhlope, 2021)



GREEN LEAFY VEGETABLES (*imifino*)

1

Tune-in

How were things in the past?



2

How are things today?



4

How can we make things more sustainable together?

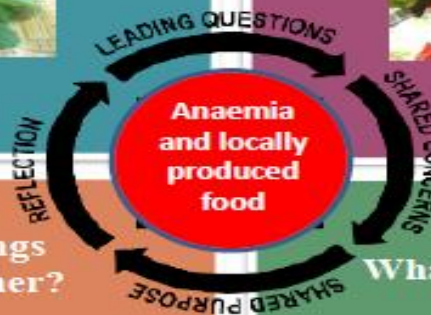
Wild-picked,

Shop-Bought,
Home-Grown
or all of the above?



3

What does this mean for us today?

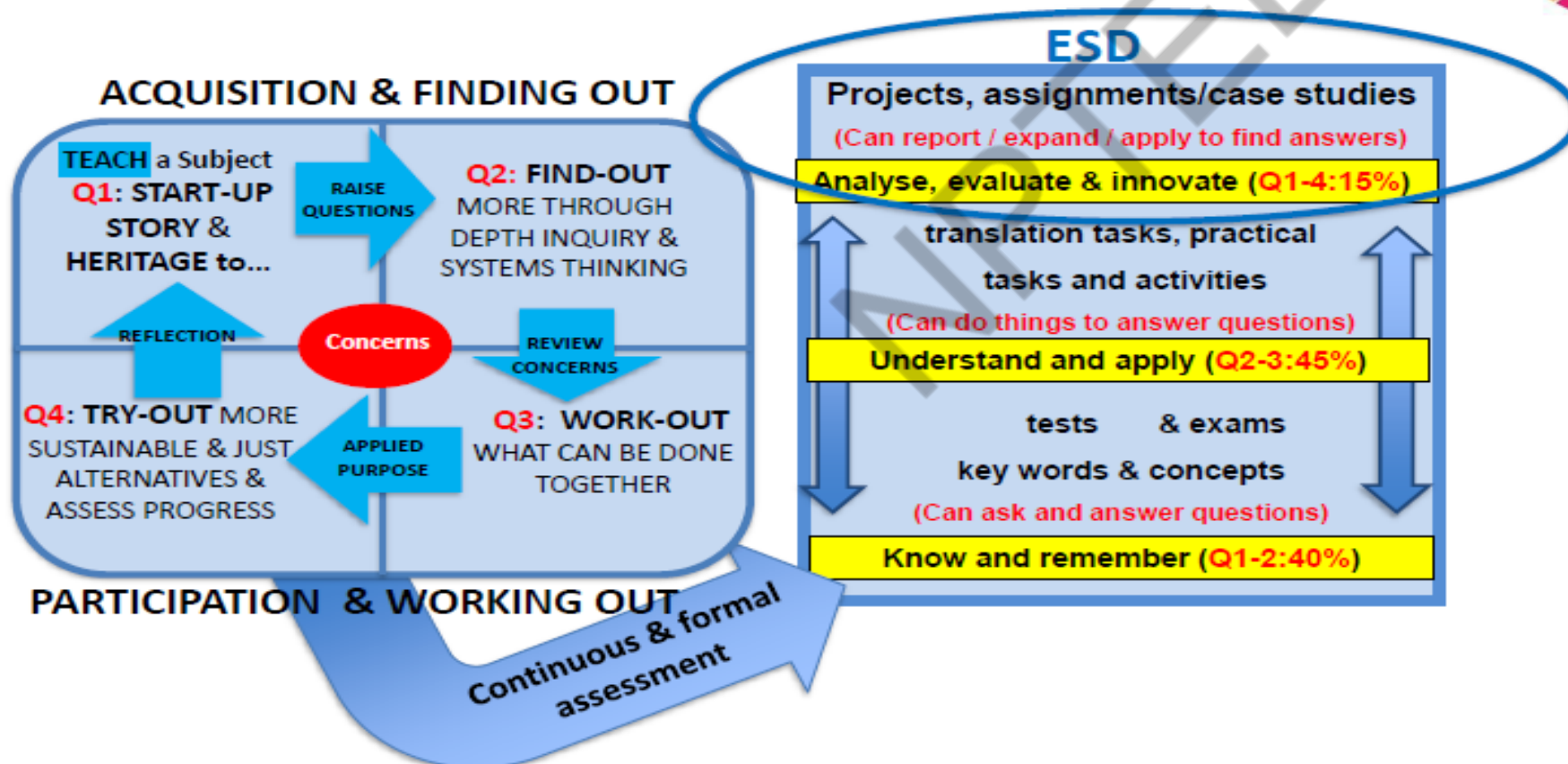


Touch the past with our memories, feel the future flying on the wings of imagination

(Mishra, 2021)



How do we assess TMESD learning with conventional assessment practices





How do we work with competence assessment for ESD in subject teaching



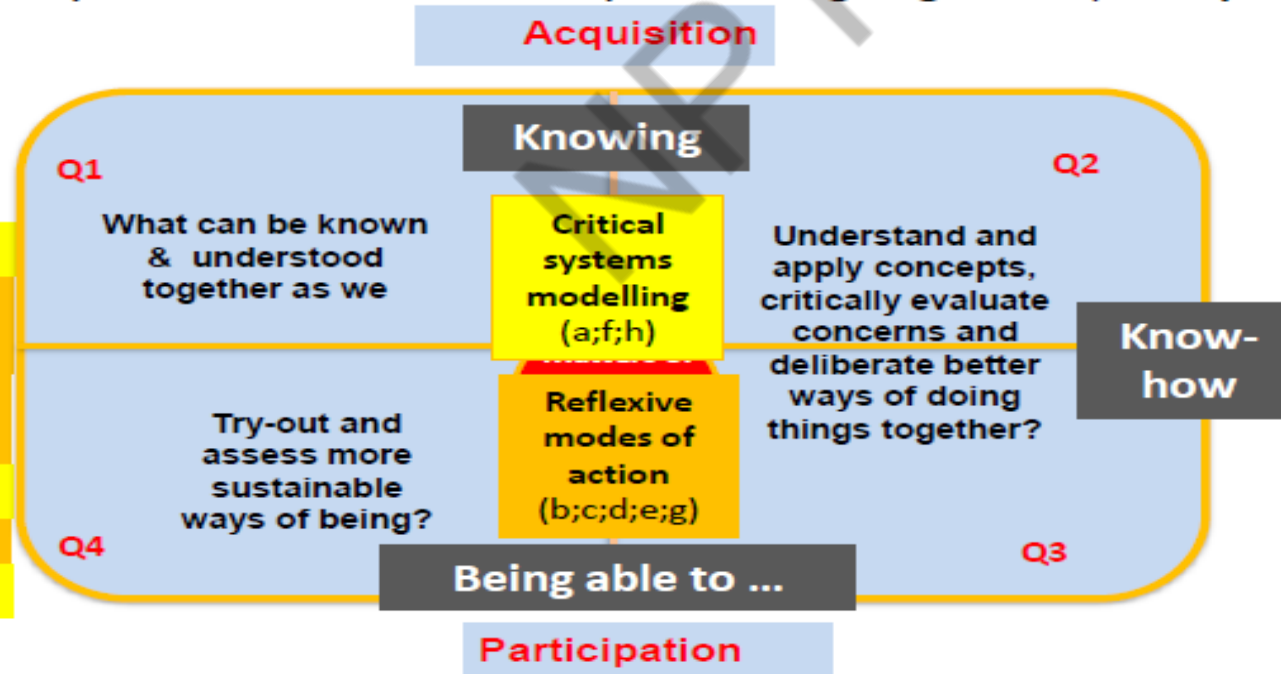
Cultural and personal competences develop with heritage and life experience as the knowledge, dispositions and aesthetics enabling us to:

- **Know things** in relation to a context and concerns with (Q1)
- Developing **Know-how** to resolve concerns and (Q2 & Q3)
- **Be able to** develop more sustainable ways of being together (Q2-4)

ESD Competences:

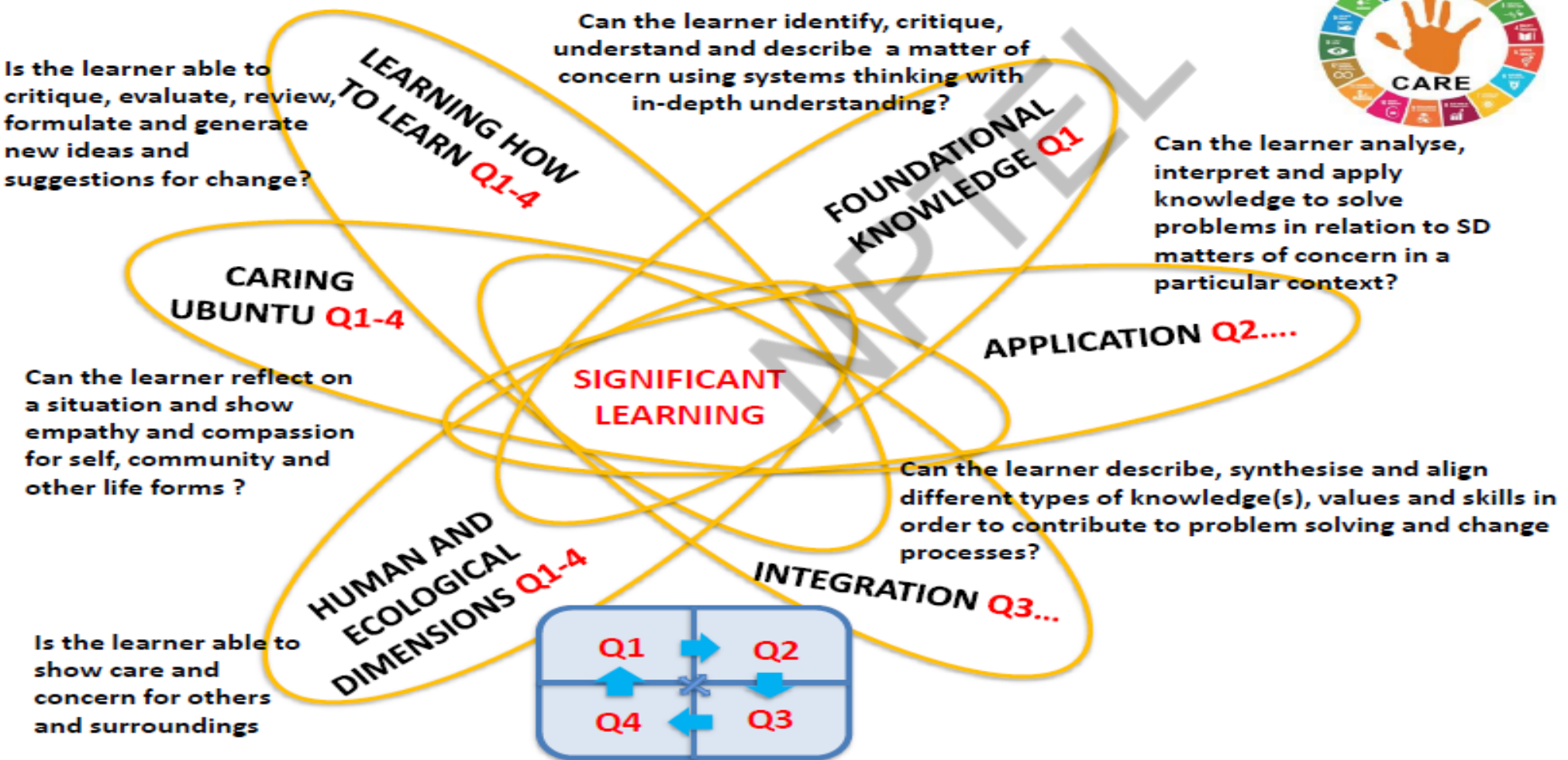
- a) Systems thinking
- b) Anticipation
- c) Normative
- d) Strategic
- e) Collaboration
- f) Critical thinking
- g) Self awareness
- h) Problem solving

(UNESCO, 2017 – p.10).





Assessing significant learning actions together





• References

- <http://www.undp.org/>
- <http://www.worldbank.org/>
- <http://www.unicef.org/>
- <http://www.imf.org/>
- <https://www.unglobalcompact.org/what-is-gc/our-work/social/poverty>
- <http://www.unesco.org/new/en/social-and-human-sciences/themes/international-migration/glossary/poverty/>
- <https://www.unisdr.org/>



LECTURE : 29

Sustainable and Clean Energy



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SDG 7 : AFFORDABLE AND CLEAN ENERGY



Objective -

Ensure access to affordable, reliable, sustainable and modern energy for all

SDG 7 has 5 major targets-

- Proposed to be accomplished by 2030
- 3 Result Targets
 - Universal admittance to present day energy
 - increment worldwide level of environmentally friendly power
 - twice as much gain in energy efficiency
- Other 2 targets are related to methods to accomplish these
 - to broaden access to sustainable energy research, innovation, and interest
 - grow and update energy administrations for non-industrial nations

AFFORDABLE &
CLEAN ENERGY



Tracking SDG 7: The Energy Progress Report (2021)

Key Findings of the Tracking SDG7: The Energy Progress Report

2010		Latest Data
1.2 billion people without access to electricity		759 million people without access to electricity (2019)
3 billion people without access to clean cooking		2.6 billion people without access to clean cooking (2019)
16.4% share of total final energy consumption from renewables		17.1% share of total final energy consumption from renewables (2018)
5.6 MJ/USD primary energy intensity		4.8 MJ/USD primary energy intensity (2018)
10.6 USD billion international financial flows to developing countries in support of clean energy		14 USD billion international financial flows to developing countries in support of clean energy (2018)



Affordable and clean energy



Challenges

ONE IN FIVE PEOPLE STILL LACK ACCESS TO MODERN ELECTRICITY and three billion people rely on wood, charcoal or animal waste for cooking and eating



Every year **INDOOR AIR POLLUTION** kills 4.3 million people – most of them women and children



Energy provided by **FOSSIL FUELS** is the **MAIN CONTRIBUTOR OF CLIMATE CHANGE** representing 60% of all greenhouse gas



Solutions

Ensure everyone has access to **CLEAN, AFFORDABLE, RELIABLE** and **MODERN ENERGY**



Invest in **RENEWABLE ENERGY** and disseminate its use

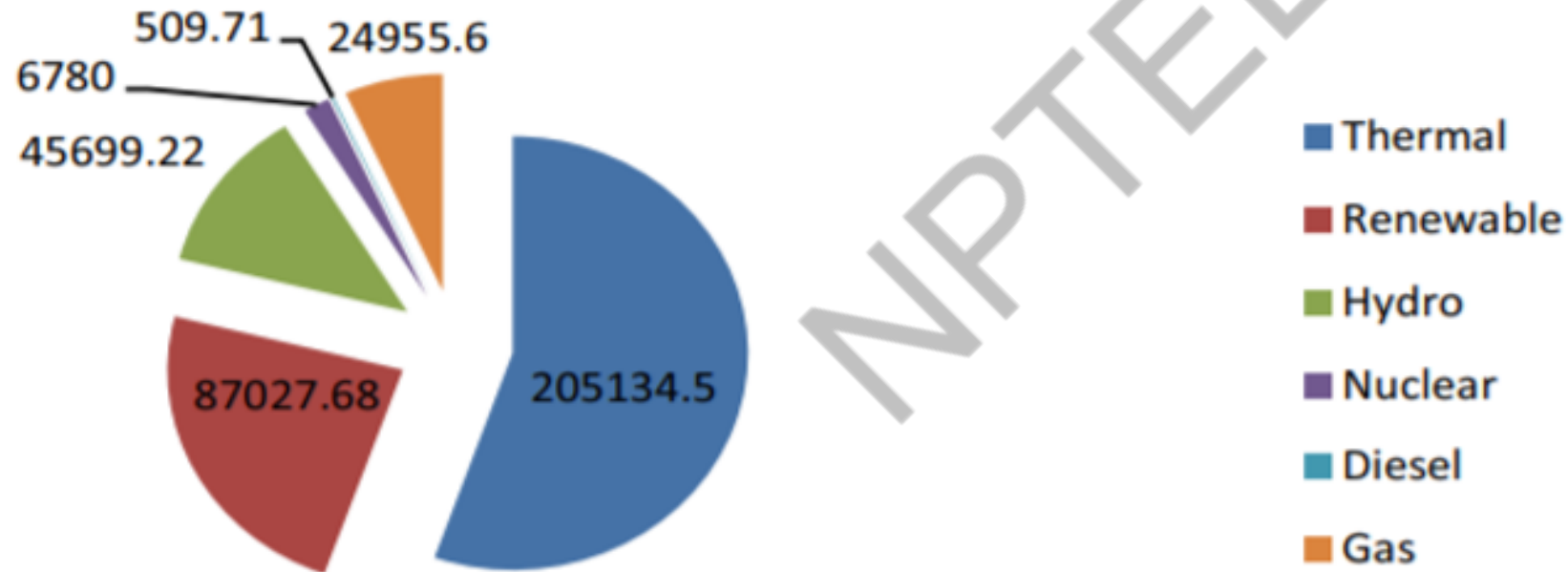


Put in place **ENERGY SAVING POLICIES**





INDIA'S SOURCE-WISE ELECTRICITY GENERATION CAPACITY IN MW



Conventional and Renewable energy Generation Capacity of India in MW (31 March 2020)



Top 5 States in installed renewable capacity

These five states have about **66.991%** of renewable energy capacity

Karnataka	17.485%
Tamil Nadu	16%
Maharashtra	12.532%
Gujarat	10.6%
Rajasthan	10.224%

Alongside these five states, Andhra Pradesh, Madhya Pradesh, Telangana and Uttar Pradesh are the other most important Indian state of installed renewable capacity. These nine states account for 91.655% of total Indian installed capacity

**AFFORDABLE &
CLEAN ENERGY**



NATION WIDE POLICIES AND TARGETS

CURRENT STATUS

CURRENT TARGET:

The Government of India has set a target of installing of installing 175 GW of renewable energy capacity by the year 2022, which includes 100 GW from solar, 60 GW from wind, 10 GW from bio-power and 5 GW from small hydro-power.

POLICY 1

This portion focuses on behaviour towards the industries that produce energy at present.

POLICY 2

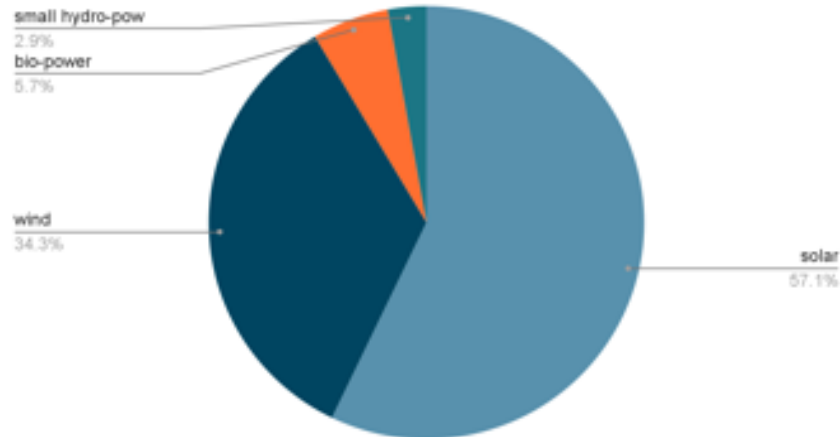
This policy addresses the future needs of energy for sustainable growth. It is necessary to address the sustainability part of sustainable energy.

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CURRENT STATUS →

TARGET 2022: RENEWABLE ENERGY SOURCES (175 MW)



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ELECTRIC VEHICLES: There are several types of Electric Vehicles, such as Battery Electric Vehicles (BEV), Zero-Emission vehicles (ZEVs) and Pure Electric Vehicle (PEV)

- The government is planning to reduce spending on petroleum product by a shift to **100% electric vehicles before 2030**.
- The Indian government has already started initiatives to provide electric vehicles on a **zero down payment** option to promote the sales of electric vehicles.
- In addition to the financial incentives, the government of India has been agreed to **sponsor up to 60% of the research and development (R&D) costs** for developing indigenous low-cost electric vehicle technology. [1]

ALTERNATE SOURCES OF ENERGY: India is set to achieve a capacity of 175 GW worth of renewable energy by the end of 2022, which expands to 450 GW by 2030 [2].

BIOENERGY: It is the single largest renewable energy source today, providing 10% of world primary energy supply.



DECENTRALISATION OF POWER

- ❖ Encouraging installation of community solar farms
- ❖ Electrifying households and areas of income-generation activity

**INCREASE
R&D FUNDS
FOR GOV.
AND
PRIVATE
COMPANIES**

POLICY 1

ENERGY DIVERSIFICATION

- ❖ Using different sources, suppliers and transportation routes

**THIS PORTION FOCUSES ON
BEHAVIOUR TOWARDS THE
INDUSTRIES THAT PRODUCE
ENERGY AT PRESENT.**

**AFFORDABLE &
CLEAN ENERGY**



POLICY 2 →

This policy addresses the future needs of energy for sustainable growth:

- Profitable scaleup of Renewables, its distribution, services and Energy Solution businesses
- Compulsory hybrid power source for all energy intensive industries
- Micro grid to every remote location for improved electrification
- Limitation to the amount of power consumed and penalty on exceeding the limit, for large scale industries



**AFFORDABLE &
CLEAN ENERGY**



The global power market is experiencing rapid power demand growth as markets recover from the pandemic. Despite all the capacity additions in renewables generation, the **amount of power currently generated by renewables is still not enough to meet this increased demand.**

The global supply of renewables **will grow by 35 gigawatts from 2021 to 2022**, but global power **demand growth will go up by 100 gigawatts** over the same period.

The world is not investing enough to meet its future energy needs. Transition-related spending is gradually picking up, but remains far short of what is **required to meet rising demand for energy services in a sustainable way.**

PITFALLS

THE GLOBAL ENERGY DEMAND IS GROWING FASTER THAN EVER

The total amount of renewable energy that's available is growing. But the increase in renewable energy is still lower than the increase in global energy demand overall.



- After having suggested the policies, it is necessary to do a Strengths, weaknesses, opportunities and threat analysis.

• SWOT ANALYSIS

 S	• Cheaper, cleaner and sustainable in the long run • Stable financial market as fuel prices fluctuate a lot • Less monopoly by a few big companies
 W	• Lack of infrastructure in India • Faster growth of energy demand • Geographical constraints
 O	• Diversification of Energy Sources • Affordable energy for all • Influence in global market and decisions
 T	• Corruption and lack of interest in the lawmakers as well as the industrialists. • Growing energy demands

**AFFORDABLE &
CLEAN ENERGY**



Targets linked to the environment:

Target 13.1: Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries

Target 13.2: Integrate climate change measures into national policies, strategies and planning

Target 13.3: Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning

From 1880 to 2012, average global temperature increased by 0.85°C.





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LECTURE : 30

Sustainable and Clean Energy



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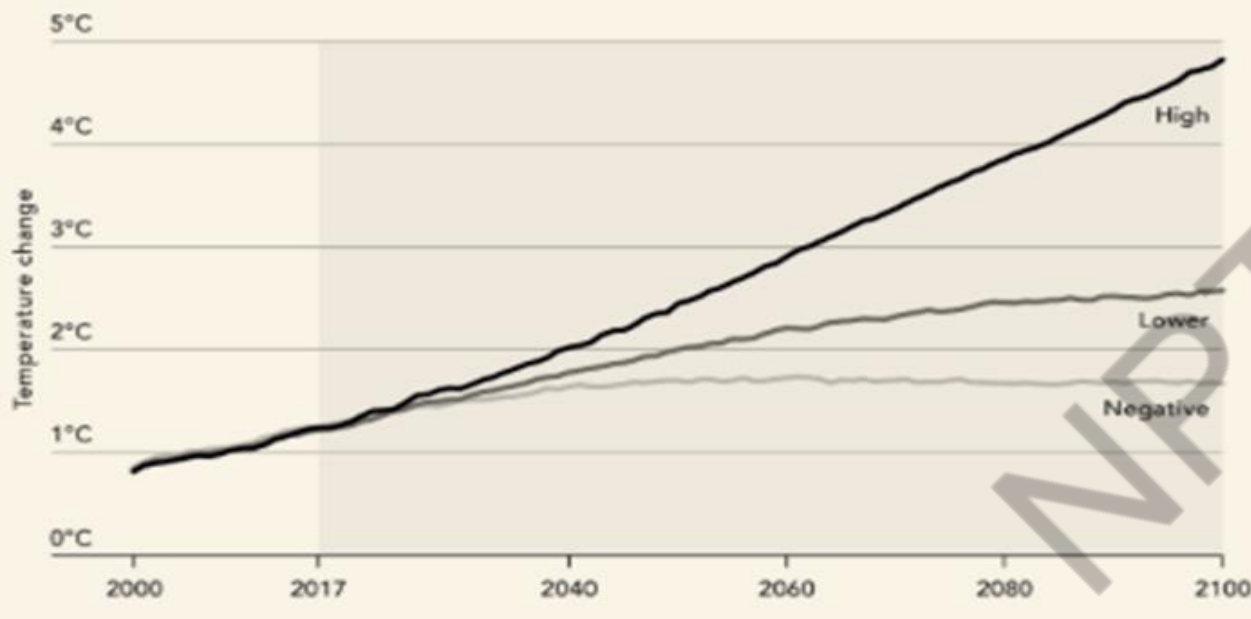


- **Target 13.a: Implement the commitment undertaken by developed-country parties to the United Nations Framework** Convention on Climate Change to a goal of mobilizing jointly \$100 billion annually by 2020 from all sources to address the needs of developing countries in the context of meaningful mitigation actions and transparency on implementation and fully operationalize the Green Climate Fund through its capitalization as soon as possible
- **Target 13.b: Promote mechanisms for raising capacity for effective climate change-related planning and management** in least developed countries and small island developing States, including focusing on women, youth and local and marginalized communities

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<https://www.bing.com/videos/search?q=climate+change+facts&&view=detail&mid=19C43F000571FE700D6C19C43F000571FE700D6C&&FORM=VDRVRV>



INCOME-ENERGY DEPENDENCE

Income and energy use go hand in hand. David MacKay showed me a chart like this one plotting energy consumption and income per person. The connection is unmistakable. (IEA; World Bank)



TEMPERATURE-EMISSION DEPENDENCE

Three lines you should know. These lines show you how much the temperature might go up in the future if emissions grow a lot (that's the high line), if they grow less (lower), and if we start removing more carbon than we emit (negative). (KNMI Climate Explorer)



TEMPERATURE LEADING TO 'KNOCK ON' EFFECTS!!!



HURRICANES



RISING SEA LEVEL



STORMS



WILDFIRES



CLIMATE DISPLACEMENT



COVID 19-CLIMATE ACTION LINK

- The **pandemic in 2020 vs climate change in 2030**
- We can determine the degree by which a worldwide pandemic affects the **global mortality rate** by averaging data from the 1918 Spanish flu and the COVID-19 pandemic over the course of a century. Every year, there are approximately 14 deaths per 100,000 individuals.
- What does this mean in terms of climate change? **Increases in global temperatures are expected to elevate worldwide mortality rates** by the same amount by the middle of the century—14 fatalities per 100,000. If emissions growth continues at its current rate, climate change could result in 75 more fatalities per 100,000 people by the end of the century.
- Therefore, climate change might be just as harmful as COVID-19 by mid-century, and five times as dangerous by 2100.
- “We now have five times the number of recorded weather disasters than we had in 1970 and they are seven times more costly. Even the most developed countries have become vulnerable”, said the UN chief.
- The Covid recovery efforts should be aligned with **national climate change and air quality strategies to reduce risks from cascading climate hazards, and gain health co-benefits.**



Evolution of International Agreements

- **Climate Action Evolution-I**
- **Sustainable??**
- The sustainable development includes-
- **sustainable economic development**
- **equitable social environment and**
- **sustainable natural and built environment**
- If one looks at the official definition it can be seen from **Brundtland report of 1987** (our common future), that outlines- meeting the needs of present without compromising future generation i.e., something on the lines of intergenerational equity.



1972 Stockholm > IPCC 1988> 1992 Rio Earth (CBDR discussion started)> 1994 UNFCCC born > 1997 Kyoto (COP3)- climate is biggest problem> 2015 Paris Agreement (NDCs, Agenda 2030) IPCC -1988 by the World Meteorological Organization (WMO) and United Nations Environment Programme (UNEP) and they underlie negotiations at the UN Climate Conference



Climate Action Evolution-II-COP

- **CoP-climate (mitigate adaptation awareness) under aegis of UNFCCC- stabilization of greenhouse gas concentrations in the atmosphere - Anthropocentric to ecocentric**
- 1995 COP 1
- 1997, the famous Kyoto Protocol
- 2002 COP 8 technology transfer focus
- 2015 COP21 limit global warming to well below 2, preferably at 1.5 degrees Celsius, compared to pre-industrial levels.'
- 2021 COP 26- resilient infra; resilient infra; \$100bn in climate finance per year by 2020 (developed); Paris Rulebook
- **The Intergovernmental Panel on Climate Change (IPCC) had won the Peace Nobel in 2007, an acknowledgement of its efforts in creating awareness for the fight against climate change.**



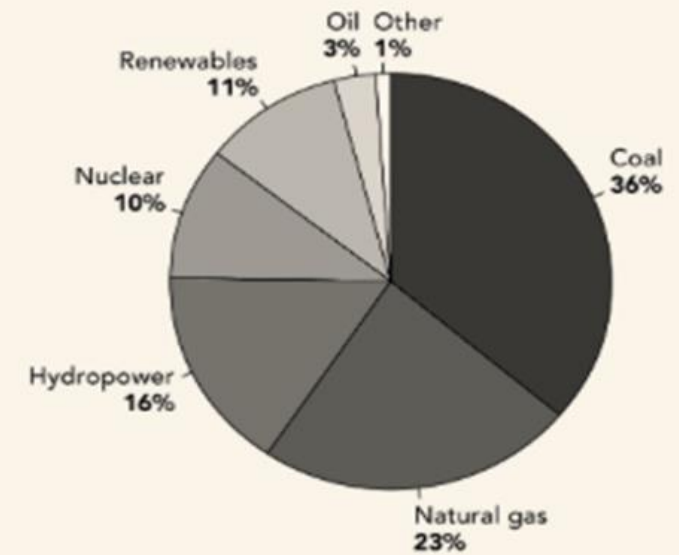
Climate Action Evolution-III -Climate Finance

- **Convention (UNFCCC)**, the Kyoto Protocol and the Paris Agreement call for financial assistance.
- In accordance with the principle of “common but differentiated responsibility and respective capabilities”
- Theme -to transfers of public resources from developed to developing countries
- **UNFCCC MANDATE** LED to
- the Global Environment Facility (GEF) has acted as the financial mechanisms operating institution.
- **Copenhagen Accord**: Parties agreed for a “goal” for the world to raise \$100 billion per year by 2020, from “a wide variety of sources”, to help developing countries cut carbon emissions (mitigation) and adaptation
- **COP 16 (2010)**: Parties established the Green Climate Fund (GCF) and in 2011 (COP 17) also designated it as an operating entity of the financial mechanism.
- Establishment of special funds: **Special Climate Change Fund (SCCF)**, the Least **Developed Countries Fund (LDCF)**, both managed by the GEF; and the Adaptation Fund (AF) under the Kyoto Protocol in 2001
- Way forward-
- **Net Zero to Net minus is India's stand for developed (and developed want India to commit to net zero).**
- **Goals to commitments should be emphasized**



LOOP: Climate change > GHG emissions> CO₂e > Reasons (electricity factor> mitigate)

- Challenges of green electricity:
- Intermittency
- Size
- Capital investment
- Risk: breaks in electricity supply
- Safety: Chernobyl Bhopal Fukushima
- Unfavorable Policies: tax rules to environmental regulations
- Outdated laws: climate change not in 1986 Acts
- election cycle
- energy innovation
- No Global cooperation
- The U.S. withdrawal from the 2015 Paris Agreement—a step that President-elect Joe Biden reversed—only illustrates that it's as hard to maintain global compacts as it is to create them in the first place.



Getting all the world's electricity from clean sources won't be easy. Today, fossil fuels account for two-thirds of all electricity generated worldwide. (bp Statistical Review of World Energy 2020)



<https://www.bing.com/videos/search?q=Sustainable+%26+clean+energy&ru=%2fvideos%2fsearch%3fq%3dSustainable%2b%2526%2bclean%2benergy%26FORM%3dHDRSC3&view=detail&mid=67991EA03A1B89ABEDB067991EA03A1B89ABEDB0&rvsmid=6593ED04E3978F29553B6593ED04E3978F29553B&FORM=VDQVAP>

RENEWABLE ENERGY AND SUSTAINABLE DEVELOPMENT

Accounting for Impacts on the Path to 100% RE





Renewable Energy Source



- Societies around the world are on the verge of a profound and urgent necessary transformation in the way they produce and use energy.
- This **shift is moving** the world away from the consumption of fossil fuels (which cause climate change and other environmental and social challenges) **toward cleaner, renewable forms of energy**.
- Despite the growing development and the interest in RES, some **economic sectors such as energy production, transportation, and industry still suffer from releasing a large number of CO₂ emissions** caused by the use of fossil fuels .
- **These emissions disturb the global climate by the rise in temperature and water level** in some areas and the increase of meteorological phenomena.

Renewable Energy contd....



- Millions of people around the world already use renewable energy to generate electricity, heat and cool buildings, cook and provide mobility. **Renewable energy is market-ready and price competitive with conventional sources** in many jurisdictions, and met about 19% of the world's final energy demand in 2014.
- Around the world, communities, islands, and cities have found that making the **transition to 100% renewable energy is largely a matter of political will** and that the **required technologies** already are at hand.
- An increasing number of governments at all levels and on **all continents are setting ambitious targets for renewable energy**, with an ever-growing number of jurisdictions aiming for 100% renewables.
- The rapid deployment of renewable energy has been driven mainly by a **wide range of objectives (drivers)**, which include advancing economic development , improving energy security, enhancing energy access and mitigating climate change.